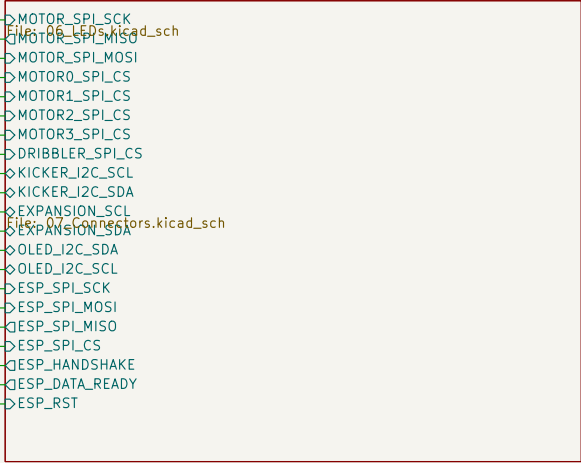
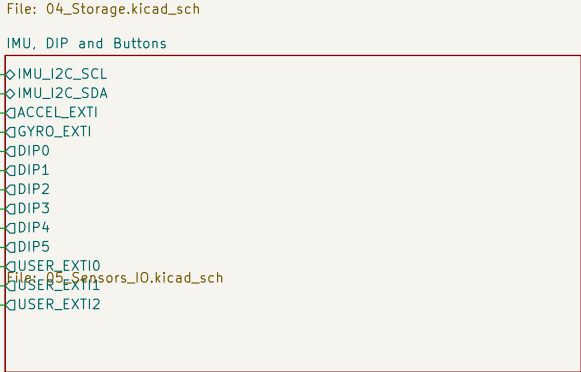
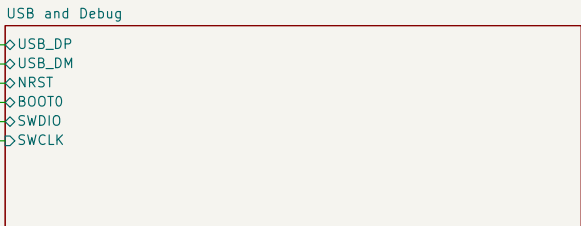
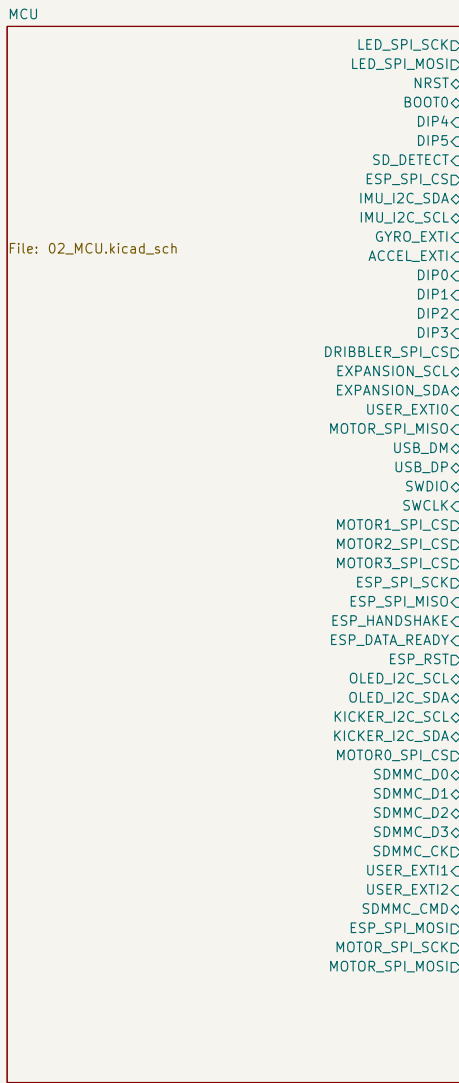
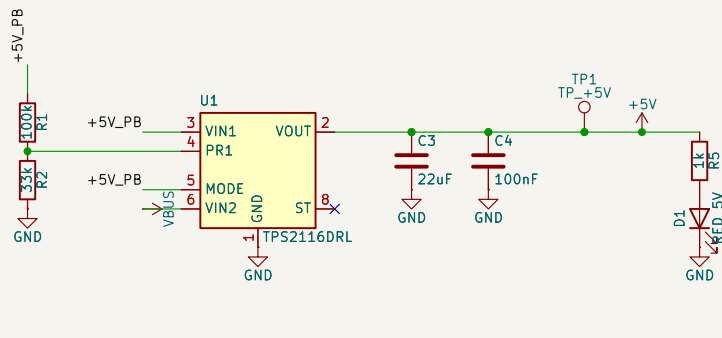
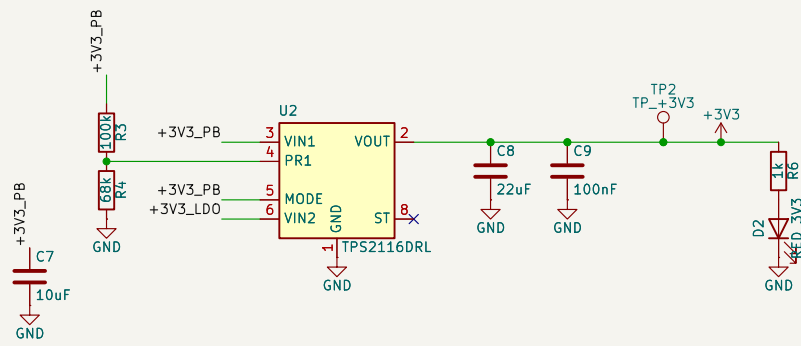




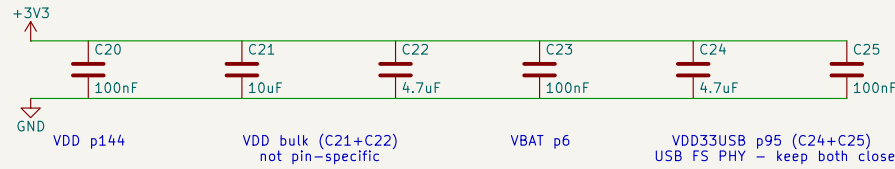
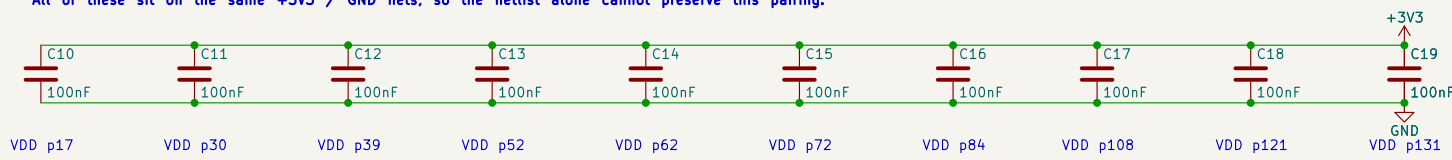


CAUTION – USB-only operation is for BENCH PROGRAMMING, not full operation: U3 (TLV1117–33) is 800 mA and dissipates $(5.0-3.3) \times 1 = 1.4 \text{ W}$ at 800 mA. The radio module alone draws 650 mA on +3V3, plus ~330 mA on +5V for LEDs/OLED. That exceeds both a default USB port and U3's comfortable dissipation.



Sheet: /Power/		
File: 01_Power.kicad_sch		
Title:		
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.0		Id: 2/8

DECOUPLING PLACEMENT – the label under each cap names the STM32 pin it must be placed beside at layout.
All of these sit on the same +3V3 / GND nets, so the netlist alone cannot preserve this pairing.



CLOCKING – no external crystal, by design.
ControlBoard_Testing_V0.2 assigns all three oscillator pins to GPIO:
PH0-OSC_IN (pin 23) = DIP4 PH1-OSC_OUT (pin 24) = DIP5 PC14-OSC32_IN (pin 8) = USER_EXTI2
So there is no HSE and no LSE. The core runs from HSI (64 MHz, +/-1%) through the PLL, and the USB DFU port uses HSI48 + CRS trimmed against USB S0F – which is exactly what the H723 provides for crystal-less USB. This follows the authoritative pin config as given.
To add real HSE/LSE later: move DIP4/DIP5 to free GPIO (PF12 pin 50 / PF13 pin 53, both free and still on port F) and USER_EXTI2 to PE1 (pin 142; EXTI1 is unused), which frees PH0/PH1 and PC14/PC15.

VREF+ (pin 32) is NOT listed in the CubeMX report but is a real pin on the LQFP144 part and must be driven, or ERC fails. It is tied to the VDDA net with 1uF + 100nF (C30/C31). FB1 isolates VDDA and VREF+ from the digital +3V3 rail; #FLG001 marks that net as driven for ERC (the ferrite is passive). PDR_ON (pin 143) tied to +3V3 = internal power-down reset enabled. VBAT (pin 6) tied to +3V3 – no backup battery is used. VCAP1/VCAP2 each take 2.2uF to GND as the H7 core LDO requires.

